

Capstone Project by CA. Rajat Bhatia

AICA - Level 2 - Batch 85

Implementation Plan

Implementation Plan - University MOU & Agreement Repository

Build a complete, production-ready, and deployable web application for a university to maintain a centralized repository of MOUs, contracts, and agreements with AI-assisted document parsing, expiry tracking, clause suggestion guidance, department isolation, and administration.

User Review Required

IMPORTANT

Key Architecture Decisions:

1. Tech Stack:

- **Backend:** Node.js + Express with SQLite (better-sqlite3 / sqlite3), multer for secure document storage, pdf-parse & mammoth for PDF/DOCX text extraction, @google/genai / Gemini API client with a built-in heuristic/regex NLP fallback engine (so the app is 100% functional out of the box even without an API key).
- **Frontend:** React (Vite) Single-Page Application with custom university-grade Design System (Navy #0F2A4A, Academic Gold #D97706, Emerald #059669, Slate #F8FAFC), Lucide icons, responsive sidebar, dynamic charts, searchable tables, side-by-side document review, and modal editors.

2. Authentication & RBAC:

- JWT-based authentication with bcrypt password hashing.
- Role-based isolation: **Admin** sees all agreements, users, departments, activity log; **Department User** strictly accesses, uploads, and edits their department's agreements.

3. AI Document Reading & Clause Suggestions:

- Automatically extracts parties, dates, purpose, financial terms, and clause summaries upon uploading PDF/DOCX.
- If a Gemini API Key is configured (via .env or in-app configuration), it uses Gemini 2.5/Flash for high-accuracy parsing and clause risk analysis.
- If no key is provided, an intelligent text-parsing rule engine extracts fields and performs legal clause gap analysis with the required legal guidance disclaimer.

4. Sample Seed Data:

- Pre-populated with authentic university departments (Computer Science, School of Law, Biotechnology, Business & Management, Office of International Relations, Research & Innovation).
- 12+ sample agreements spanning Active, Expiring Soon, Expired, Financial & Non-Financial with generated sample PDF/DOCX documents.